

Digital Material Passport

ID **DMP-METAL-004** - Version **1.0.0**
Issue Date **2025-05-16** - Certificate Type **EN 10204 3.1**

Manufacturer		Customer	Business Transaction									
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Advanced Nuclear Systems Ltd. Energy Square 456 Abingdon OX14 3DB GB procurement@advnuclear.example.com	Order	PO-56789 · 200 kg Pos. 2 · 2025-04-10								
			Delivery	DN-12345 · 200 kg Pos. 1 · 2025-05-15 Internal Order ID: SO-98765								
Product												
Product Name	Stainless Steel 316L		Number (UNS)	S31603								
Batch ID	H-87654-01		Name (EN)	1.4404								
Surface Condition	2B		Shape	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm								
Weight	200 kg		Marking	316L stamped on corner with medium depth and good legibility.								
Production Date	2025-05-14			Coloring	Natural finish with full coverage for protection.							
Country of Origin	DE		Product Norm		ASTM A240 (2023)							
Chemical Analysis												
Heat Number H-87654 - Melting Process EAF+AOD+LF - Casting Method ContinuousCasting - Casting Date 2025-05-13 - Sample Location Ladle												
Symbol	C	Cr	Ni	Mo	Mn	Si	P	S	N			
Unit	%	%	%	%	%	%	%	%	%			
Min		16.0	10.0	2.00								
Max	0.030	18.0	14.0	3.00	2.0	0.75	0.045	0.030	0.100			
Actual	0.018	17.2	10.5	2.15	1.4	0.38	0.025	0.002	0.052			
Mechanical Properties												
Tensile Strength ¹		Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
Value [MPa]		Rm	578	580	582	580	578	582	515	ASTM E8	✓	
0.2% Yield Strength ¹		Symbol	1	2	3	Average	Min	Max	Std Dev	Spec Min	Method	Status
Value [MPa]		Rp0.2	238	240	242	240	238	242	2	205	ASTM E8	✓
Elongation ¹		Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
Value [%]		A	51	52	53	52	51	53	40	ASTM E8	✓	
¹ 3 specimens tested												
Supplementary Tests												
Test		Result / limits					Method		Status			
Intergranular Corrosion Resistance		Yes - No evidence of intergranular attack					ASTM A262 Practice E		✓			
Pitting Corrosion Resistance 72 hours at 22°C in 6% FeCl ₃		1.2 g/m² - max 4.0 g/m²					ASTM G48 Method A		✓			
Crevice Corrosion Resistance 72 hours in 3.5% NaCl solution		Yes - No visible crevice corrosion					ASTM G78		✓			
Stress Corrosion Cracking Resistance Boiling 42% MgCl ₂ solution, 100 hours		Yes - No cracking observed					ASTM G36		✓			
Ferrite Content		2.5 % - max 5.0 %					ASTM A800		✓			
Grain Size		7 ASTM No. - min 5 ASTM No.					ASTM E112		✓			
Inclusion Rating Worst field rating		A1, B1, C1, D1 - max A2, B2, C2, D2					ASTM E45 Method A		✓			
Ultrasonic Examination		Yes - No recordable indications					ASTM A388		✓			
Liquid Penetrant Examination		Yes - No relevant indications					ASTM E165		✓			
Weldability Test		0.4 mm - max 1.0 mm					Varestraint Test		✓			
Surface Finish		25 µin Ra - max 32 µin Ra					ASME BPE SF1		✓			

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PREN (Pitting Resistance Equivalent Number) Calculated using formula: %Cr + 3.3 × %Mo + 16 × %N	25.8 - min 24.0		✓
Dimensional Tolerance	-0.3 - 0.2 mm - target -0.4 - 0.4 mm	ASTM A480	✓
Flatness	4 mm/m - max 9 mm/m	ASTM A480	✓
PMI (Positive Material Identification)	Yes - Material confirmed as 316L stainless steel	XRF Analysis	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM A240/A240M and meets all specified requirements. This material is suitable for nuclear applications in accordance with RCC-M code.

Validated by **Thomas Wagner**, Metallurgist, Quality Assurance - 2025-05-16
Validated by **Anna Schmidt**, Quality Manager, Quality Assurance - 2025-05-16